

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2022-2023
FULL MARKS: 75

CSE 4643: Mobile Application Development

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Two childhood friends Atiq and Shafin have met after a long time. They realize that both of them are web developers. Shafin develops with React and Atiq develops with classic HTML, JS, CSS approach. Now after reminiscing, they started talking about the present. Shafin says classic approaches with HTML, JS, and CSS are the best but Atiq says React is the best. Now these two friends approached you for the answer. Please give them the answer with proper justification. 8
(CO4)
(PO2)

- b) Differentiate between VAR and LET keywords using the following Code Snippet 1, and visualize the dissimilarities with stack procedure. 8
(CO1)
(PO1)

```
1 var v1 = 10;
2 let l2 = 17;
3 const f1 = () => {
4     var v0 = 9;
5     let l0 = 8;
6     if(1){
7         let l1 = 7;
8         var v3 = 121;
9         console.log(l0);
10        console.log(l1);
11        console.log(v1);
12        console.log(l2);
13    }
14    //console.log(l1);
15    console.log(v3);
16 };
17 f1();
18 console.log(v0);
```

Code Snippet 1: Code for question 1(b)

- c) Edon, the JavaScript developer, wants to teach his juniors about the output methods, but he only knows the console.log() method. Help him with other methods. 4
(CO1)
(PO1)

- d) Find the output of the following Code Snippet 2. 5
(CO2)
(PO1)

```
1 const names = ['atiq', 'shafin', 'rahman', 'edon', 'jafrin']
2
3 console.log(names.filter(nm => nm.length < 6))
```

Code Snippet 2: Code for question 1(d)

2. a) ATQDevelopers is a startup software company that primarily develops mobile applications. The CEO of this company got many complaints from the clients that the navigation system of the developed applications is confusing and complex. Now they are working on a new project and the CEO doesn't want to get the complaints this time, so he hired you as a mobile design pattern expert. Illustrate a navigation pattern that is best for the following use case with proper justification: 10
(CO4)
(PO2)
- IUTPS is the client for this new project and they want to launch a new mobile application to showcase their photos. This application will have a few primary categories which are SHOOT-AT-IUT, Contests, and Winners, and these categories are equivalent in hierarchy. All the categories should be always visible for quick access. A unique feature asked by IUTPS is that the title SHOOT-AT-IUT might change to something else like SHOOT-ON-16December depending on the current contest they are arranging. According to the IUTPS president, they intend to update the contents regularly. Since it is to showcase the photos, the application's UI is bound to be simple and efficient.
- b) Find the anti-patterns and discuss respective solutions from the following scenario: 312Technology a well-known software developer company built an application called Medication Tracker which will notify users when they should take their medications. The developers used a side drawer navigation system positioned on the bottom. They added lots of primary categories with varying importance. The most important features are on the top but sometimes users might need to scroll to access the feature they want. 5
(CO3)
(PO2)
- c) Discuss two reasons sidebars will never be considered a persistent navigation pattern. 5
(CO1)
(PO1)
- d) Define and illustrate the Skeuomorphic pattern. 5
(CO1)
(PO1)
3. a) Find the outputs of the following Code Snippet 3 for two cases where case 1: id equals 8 and case 2: id is equal to 27. 5
(CO2)
(PO1)
- ```

1 switch(true){
2 case (id%2)===0:
3 console.log('id is even');
4 break;
5 case (id%3)===0:
6 console.log('id is multiple of 3');
7 case (id%2) !==0:
8 console.log('id is odd');
9 break;
10 default:
11 console.log('id is nothing of the above');
12 }

```
- Code Snippet 3:** Code for question 3(a)
- b) Replace ??? with proper keywords of react native so that the following Code Snippet 4 will render the following in Figure 1. 5  
(CO2)  
(PO1)



**Figure 1:** An image of a component for question 3(b)

```

1 import {Text,View,StyleSheet} from 'react-native'
2 import {useEffect,useState} from '????'
3 const CalculateExpense = ({expData}) => {
4
5 const [es,setEs] = useState(0)
6 const [pe,setPe] = useState(0)
7
8 ????(()=>{
9 if (expData && expData.length > 0){
10 let mn = new Date(expData[0].dt)
11 let mx = new Date(expData[0].dt)
12 let te = 0
13 expData.forEach((item,index)=>{
14 console.log(mn,mx,item.dt)
15 mn = Math.min(mn,new Date(item.dt))
16 mx = Math.max(mx,new Date(item.dt))
17
18 te += parseInt(item.exp)
19 })
20 setEs(te)
21 setPe((te / ((Math.abs(mx - mn) / (1000 * 60 * 60 * 24))+1)).
22 toFixed(2))
23 } else {
24 setEs(0)
25 setPe(0)
26 }
27 },[expData])
28 return (
29 <View style={styles.timerContainer}>
30 <Text style={styles.elapsedTime}>???? : {es}</Text>
31 <Text style={styles.title}>Per Day Expense : {pe}</Text>
32 </????>
33)
34 }
35 export default ????

```

**Code Snippet 4:** Code for question 3(b)

c) Discuss the following three terms with examples,

1. Asynchronous Programming
2. Setup function of useEffect
3. Conditional Rendering of React Native

6  
(CO1)  
(PO1)

d) What will be the output of the following Code Snippet 5,

```
1 const numbers = [45,4,9,16,25];
2
3 let txt = '';
4 numbers.forEach(myFunction);
5
6 function myFunction(value) {
7 txt += value;
8 }
```

4  
(CO2)  
(PO1)

**Code Snippet 5:** Code for question 3(d)

e) Differentiate between states and props.

5  
(CO1)  
(PO1)